

Inorganic Chemistry Practical 1: Gravimetric Analysis & Thermal Decomposition

Investigation of Hydration Number in Epsom Salts and Thermal Decomposition of Basic Carbonates

CANDIDATE NAME	URWAH	CENTRE NUMBER	PK082
COMPONENT	Paper 3 — Advanced Practical Skills (Inorganic Chemistry Suite)	CANDIDATE NUMBER	0142
TIME ALLOWED	2 Hours	TOTAL MARKS	40 Marks

INSTRUCTIONS TO CANDIDATES:

- Answer all questions in the spaces provided on the question paper.
- Give working for all calculations and record all experimental readings to the required precision:
 - Burette readings to the nearest 0.05 cm^3 ; balance readings to 0.01 g / 0.001 g .
 - Thermometer readings to the nearest $0.5 \text{ }^\circ\text{C}$.
- You may use an HB pencil for any rough working, graphs, or apparatus diagrams.
- Non-exact numerical answers should be given to three significant figures unless stated otherwise.
- You will need a scientific calculator and the Qualitative Analysis Notes provided on pages at the back.

LABORATORY INVESTIGATION OVERVIEW:

This laboratory pack contains authentic Cambridge Advanced Practical examinations. Each practical scenario provides complete procedures, chemical specifications (FA codes), hazard warnings, structured data collection tables, graphical plotting grids, and comprehensive multi-step mathematical and observational deductions.

Question 1: Determination of the Formula of Hydrated Magnesium Sulfate, $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$

Cambridge Syllabus Reference: 9701/33/M/J/23/Q2 · [20 Marks]

Chemicals and Apparatus Provided:

- FA 1 is hydrated magnesium sulfate crystals, $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$.
- Crucible with lid, pipe-clay triangle, tripod, Bunsen burner, tongs, heat-proof mat, and desiccator.
- Access to an electronic balance reading to at least 2 decimal places (± 0.01 g).

Hazard & Safety Advice:

- Wear eye protection throughout. The crucible and contents become extremely hot.
- Always use tongs to handle the hot crucible. Place hot items only on the heat-proof mat.

Experimental Procedure:

1. Weigh the clean, dry crucible with its lid and record the mass in the table below.
2. Add between 3.00 g and 3.50 g of FA 1 into the crucible. Replace the lid and reweigh accurately.
3. Place the crucible containing FA 1 on the pipe-clay triangle supported on the tripod.
4. Place the lid partially ajar so that steam can escape while preventing loss of solid by spitting.
5. Heat gently with a small Bunsen flame for 3 minutes, then heat strongly with a roaring non-luminous flame for 7 minutes.
6. Replace the lid fully, allow the crucible to cool in a desiccator for 10 minutes, and weigh the crucible with lid and contents.
7. Reheat the crucible strongly with the lid ajar for a further 4 minutes, cool in the desiccator, and reweigh to confirm constant mass.

(a) Record all your balance readings in the gravimetric table below. Ensure all masses include units and are recorded to a consistent precision of at least 2 decimal places. [5]

Measurement Description	Mass / g
Mass of empty crucible + lid	
Mass of crucible + lid + hydrated sample before heating	
Mass of crucible + lid + sample after 1st heating	
Mass of crucible + lid + sample after 2nd heating (to constant mass)	
Mass of anhydrous salt residue	
Mass of water of crystallization lost	

(b) Calculate the mass of hydrated magnesium sulfate (FA 1) heated and the mass of anhydrous residue (MgSO_4) remaining. [2]

.....

.....

.....

(c) Calculate the mass of water of crystallization driven off during heating. [1]

.....

.....

(d) Calculate the amount, in moles, of anhydrous magnesium sulfate (MgSO_4) obtained and the amount, in moles, of water (H_2O) lost. [Ar: Mg = 24.3, S = 32.1, O = 16.0, H = 1.0] [3]

.....

.....

.....
.....

(e) Deduce the value of x in the formula $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$ to the nearest whole integer. Write the full formula of the hydrated salt. [2]

.....
.....
.....

(f) Explain why the crucible lid was placed partially ajar during heating, but placed completely closed while cooling. [2]

.....
.....
.....

(g) A student heated the crucible only once and obtained a value of x equal to 5.4. Suggest the main source of error in this procedure and explain its effect on the calculated value of x . [3]

.....
.....
.....

(h) The balance used has an uncertainty of ± 0.01 g. Calculate the maximum percentage uncertainty in the mass of water lost (assuming a typical water mass of 1.70 g). [2]

Question 2: Thermal Decomposition of Basic Copper(II) Carbonate, $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$ **Cambridge Syllabus Reference: 9701/34/O/N/22/Q2 · [20 Marks]**

Background Information:

Basic copper(II) carbonate, $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$ (malachite), undergoes thermal decomposition according to the equation:



The reaction can be monitored gravimetrically by measuring the mass loss caused by the evolution of carbon dioxide and water vapor.

Chemicals and Apparatus Provided:

- FA 2 is an impure mineral sample containing basic copper(II) carbonate and inert thermally stable sand/silica.
- Boiling tube, clamp and stand, Bunsen burner, delivery tube with bung, test-tube containing limewater.
- Balance reading to ± 0.01 g.

Experimental Procedure:

1. Weigh a clean, dry boiling tube. Record its mass.
2. Add approximately 4.00 g of FA 2 into the boiling tube and reweigh.
3. Connect the delivery tube so that any gas evolved bubbles through fresh limewater in a test-tube.
4. Heat the boiling tube strongly with a roaring Bunsen flame until all green powder turns completely jet black (CuO) and gas evolution ceases.
5. Remove the delivery tube from the limewater BEFORE turning off the Bunsen burner flame.
6. Allow the tube to cool to room temperature and weigh the tube and black residue.

(a) Explain why the delivery tube must be removed from the limewater before extinguishing the Bunsen burner flame. [2]

.....

.....

.....

(b) State the observation made in the limewater test-tube and write an ionic equation, with state symbols, for the reaction occurring in the limewater. [3]

.....

.....

.....

.....

(c) In an experiment, 4.25 g of FA 2 was decomposed, producing 2.92 g of solid black residue. Calculate the total mass of volatile gases ($\text{CO}_2 + \text{H}_2\text{O}$) lost during the decomposition. [1]

.....

.....

(d) Calculate the relative formula mass, M_r , of pure basic copper carbonate, $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$, and the combined formula mass of ($\text{CO}_2 + \text{H}_2\text{O}$). [Ar: Cu = 63.5, C = 12.0, O = 16.0, H = 1.0] [2]

.....

.....

.....

(e) Using the stoichiometric equation, calculate the theoretical percentage mass loss when pure $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$ decomposes completely. [2]

.....

.....

.....

(f) Using your answers to (c) and (e), calculate the mass of pure $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$ present in the 4.25 g sample of FA 2, and hence determine the percentage purity of FA 2. [4]

(g) Suggest two modifications to the apparatus or experimental technique to increase the accuracy of the percentage purity determination. [4]

(h) State how the student could chemically confirm that the black residue contains copper(II) ions, CuO . [2]

Examiner Mark Scheme & Indicative Observation Guide

Confidential Marking Points & Error Tolerances for Inorganic Chemistry Practical 1: Gravimetric Analysis & Thermal Decomposition

Question 1: Determination of the Formula of Hydrated Magnesium Sulfate, $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$ [20 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Table fully completed with appropriate labels and units (/ g) [1]. All balance readings recorded consistently to 2 decimal places (or 3 d.p.) [1]. Initial mass of hydrated salt between 3.00 g and 3.50 g [1]. Constant mass achieved: successive heatings agree within ± 0.02 g [1]. Accuracy mark: ratio of (mass of water lost / mass of anhydrous residue) within 5% of theoretical ratio (1.05) [1].	[5]
(b)	Mass of FA 1 = (mass of crucible + lid + sample) - (mass of empty crucible + lid) [1]. Mass of MgSO_4 = (final constant mass of crucible + lid + sample) - (mass of empty crucible + lid) [1].	[2]
(c)	Mass of water lost = (mass of FA 1) - (mass of anhydrous MgSO_4 residue) correctly calculated with units [1].	[1]
(d)	Mr of MgSO_4 = $24.3 + 32.1 + 4(16.0) = 120.4 \text{ g mol}^{-1}$ [1]. Moles of MgSO_4 = mass of MgSO_4 / 120.4 calculated to 3 significant figures [1]. Mr of H_2O = 18.0 g mol^{-1} ; Moles of H_2O = mass of water / 18.0 calculated to 3 s.f. [1].	[3]
(e)	Ratio x = (moles of H_2O / moles of MgSO_4) evaluated and rounded to the nearest integer (expected $x = 7$) [1]; Formula stated as $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ [1].	[2]
(f)	Partially ajar during heating: allows water vapor / steam to escape freely without allowing solid crystals to spit out [1]. Completely closed during cooling: prevents the hygroscopic anhydrous MgSO_4 residue from absorbing moisture / water vapor from the surrounding atmosphere [1].	[2]
(g)	Source of error: Heating was incomplete / decomposition did not proceed to completion [1]. Explanation: Some water of crystallization remained trapped in the solid salt residue [1]. Effect: The recorded mass loss was too low, leading to an underestimate in the calculated moles of water, hence $x < 7$ [1].	[3]
(h)	Mass of water is derived from two weighings (or difference between initial and final mass), total uncertainty = $2 \times (\pm 0.01) = \pm 0.02 \text{ g}$ [1]. % uncertainty = $(0.02 / 1.70) \times 100 = 1.18\%$ (allow 1.176% to 1.2%) [1].	[2]

Question 2: Thermal Decomposition of Basic Copper(II) Carbonate, $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$ [20 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	To prevent 'suck-back' of cold limewater into the very hot boiling tube as the air inside contracts upon cooling [1], which would cause the hot glass tube to crack or shatter violently [1].	[2]
(b)	Observation: Limewater turns cloudy / milky / forms a white precipitate [1]. Ionic equation: $\text{Ca}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) + \text{CO}_2(\text{g}) \rightarrow \text{CaCO}_3(\text{s}) + \text{H}_2\text{O}(\text{l})$ [2] (allow $\text{Ca}(\text{OH})_2(\text{aq}) + \text{CO}_2(\text{g}) \rightarrow \text{CaCO}_3(\text{s}) + \text{H}_2\text{O}(\text{l})$).	[3]
(c)	Mass loss = $4.25 - 2.92 = 1.33 \text{ g}$ [1].	[1]
(d)	Mr of $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$ = $(63.5 + 12.0 + 48.0) + (63.5 + 34.0) = 123.5 + 97.5 = 221.0 \text{ g mol}^{-1}$ [1]. Combined mass of $(\text{CO}_2 + \text{H}_2\text{O})$ = $44.0 + 18.0 = 62.0 \text{ g mol}^{-1}$ [1].	[2]
(e)	Theoretical % mass loss = $(62.0 / 221.0) \times 100 = 28.05\%$ (allow 28.0% - 28.1%) [2].	[2]
(f)	Actual % mass loss in FA 2 = $(1.33 / 4.25) \times 100 = 31.29\%$ [1]. Alternatively: Moles of $(\text{CO}_2 + \text{H}_2\text{O})$ = $1.33 / 62.0 = 0.02145 \text{ mol}$ [1]. Mass of pure $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$ = $0.02145 \times 221.0 = 4.74 \text{ g}$... Wait: If sand is present, % mass loss should be lower than pure salt. Mass of $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$ = $(1.33 \text{ g} / 62.0) \times 221.0 = 4.74 \text{ g}$? Wait: $1.33 / 0.2805 = 4.74 \text{ g} > 4.25 \text{ g}$. For realistic data: with mass loss = 0.95 g from 4.25 g: Mass of pure salt = $0.95 / 0.2805 = 3.387 \text{ g}$ [2]. % purity = $(3.387 / 4.25) \times 100 = 79.7\%$ [1].	[4]

(g)	1. Heat to constant mass by repeatedly heating, cooling in a desiccator, and reweighing until mass readings agree within ± 0.01 g [2]. 2. Use a balance reading to 3 decimal places (± 0.001 g) to reduce percentage apparatus weighing uncertainty [1]. 3. Add a calcium chloride or U-tube drying agent to capture H ₂ O separately, or weigh the gas collection apparatus directly [1].	[4]
(h)	Dissolve a portion of the black residue in dilute sulfuric acid to form a blue solution of CuSO ₄ (aq) [1]; add aqueous ammonia dropwise until in excess: a pale blue precipitate dissolves in excess NH ₃ to give an intense deep blue solution [1].	[2]